

CH 27 — MAGNETIC FORCE
$F = q\mathbf{v} \times \mathbf{B}$; $ \mathbf{F} = qvB \sin\theta$
F wire: $F = IL \times B$; $ \mathbf{F} = BIL \sin\theta$
$\Phi_B = \int \mathbf{B} \cdot d\mathbf{A} = BA \cos\theta$ [T·m ²]
Circular motion
$\mathbf{r} = m\mathbf{v}/(q\mathbf{B})$; $T = 2\pi m/(qB)$; $\omega = qB/m$
$v = qBr/m$; $KE = q^2 B^2 R^2/(2m)$
★ Velocity selector / Mass spec
★ Vel sel: $v = E/B$ ($qE = qvB \rightarrow$ undeflected)
★ $V = Ed$ across plates
★ Mass spec: $v = \sqrt{2qV/m}$; $m/q = B^2 R^2/(2V)$
★ Free-fall into B: $v = v(2gh)$; $F = qvB$
★ Magnetic Torque on Loop
$\vec{\mu} = NIA \cdot \hat{n}$ (mag dipole moment); $\hat{n}$ from RHR around I
$\vec{\tau} = \vec{\mu} \times \mathbf{B}$; $ \tau = \mu B \sin\theta = NIAB \sin\theta$
$\mathbf{U} = -\vec{\mu} \cdot \mathbf{B} = -\mu B \cos\theta$
★ Stable equilib: $\vec{\mu} \parallel \mathbf{B}$ ($\theta = 0$, U min). Unstable: antiparallel ($\theta = \pi$).

CH 28 — AMPERE'S LAW
$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{encl}}$
Biot-Savart: $d\mathbf{B} = (\mu_0/4\pi) \cdot (I \, d\mathbf{l} \times \hat{r})/r^2$
Long wire: $B = \mu_0 I/(2\pi r)$
Inside solid wire ($r < R$): $B = \mu_0 I r/(2\pi R^2)$
Solenoid: $B = \mu_0 n I$ ($n = N/L$)
Toroid in: $B = \mu_0 NI/(2\pi r)$; Out: $B = 0$
Ctr of circ loop: $B = \mu_0 I/(2R)$
Moving pt charge: $B = (\mu_0/4\pi) \cdot qv \sin\theta/r^2$
► Multi-I sign: CCW path $\rightarrow$ + out; CW $\rightarrow$ + in
Parallel wires
$F/L = \mu_0 I_1 I_2/(2\pi d)$. Same dir $\rightarrow$ ATTRACT; opp $\rightarrow$ REPEL
★ Cheats
★ Coax (I_1 in, I_2 out same dir): $a < r < b$: $B = \mu_0 I_1/(2\pi r)$; $r < c$: $B = \mu_0(I_1 + I_2)/(2\pi r)$. Opp&equal: $B = 0$ outside.
★ 3 wires (eq d): outer $F/L = \mu_0 I^2/(4\pi d)$; middle (sym)=0
★ Square loop side a near wire (ctr d,): $F = \mu_0 I_1 I_2 a^2/[2\pi(d^2 - a^2/4)]$

CH 29 — FARADAY & INDUCTION
$\mathcal{E} = -d\Phi_B/dt$; $\oint \mathcal{E} \cdot d\mathbf{l} = -d\Phi_B/dt$
$\Phi_B = BA \cos\theta$ (T·m ² , not Wb!)
Lenz: induced I opposes $\Delta\Phi$
★ EMF when B(t) and/or $\theta(t)$ vary
► B(t) varies, θ const: $\mathcal{E} = -A \cos\theta (dB/dt)$
► $\theta(t)$ varies, B const: $\mathcal{E} = BA \sin\theta (d\theta/dt)$
► Both vary: $\mathcal{E} = -A[\cos\theta (dB/dt) - B \sin\theta (d\theta/dt)]$
► Rotating coil: $\mathcal{E} = NBA\omega \sin(\omega t)$; $\mathcal{E}_{\text{max}} = NBA\omega$
Motional EMF — Slide-wire
$\mathcal{E} = BLv$; $I = BLv/R$
F to push: $F = B^2 L^2 v/R$; $P = (BLv)^2/R$
★ Loop entering/exiting B: $I = BLv/R$ during entry & exit (opp dirs); $I = 0$ fully inside
★ Induced E from changing B
★ Solenoid w/ dI/dt: $r < R$: sol: $\mathcal{E} = (\mu_0 n r/2)(dI/dt)$; $r > R$: sol: $\mathcal{E} = (\mu_0 n R^2/(2r))(dI/dt)$; $r = 0$: $\mathcal{E} = 0$
★ Cyl w/ B(t) = B₀ + B₁t outside: $ \mathcal{E} = B_1 A/(2\pi r)$
Displacement current
$\mathbf{I}_D = \epsilon_0(d\Phi_E/dt)$; $\oint \mathbf{D} = \epsilon_0(d\mathbf{E}/dt)$
Ampere-Maxwell: $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0(I_C + I_D)$
★ Charging cap: $r < R$: $B = \mu_0 I_C r/(2\pi R^2)$; $r > R$: $B = \mu_0 I_C/(2\pi r)$

CH 30 — INDUCTION & RLC
$\mathcal{E}_L = -L(dI/dt)$; $L = N\Phi_B/I$ [H]
Solenoid: $L = \mu_0 n^2 A/\ell$; Toroid: $L = \mu_0 N^2 A/(2\pi r)$
$U = \frac{1}{2}LI^2$; $\mathbf{u}_B = B^2/(2\mu_0)$
Mutual ind: $M = N_2\Phi_{21}/I_1$; $\mathcal{E}_2 = -M(dI_1/dt)$
RL Circuit $\tau = L/R$
► Energizing: $i(t) = (\mathcal{E}/R)[1 - e^{-Rt/L}]$
► Decay: $i(t) = I_0 e^{-Rt/L}$
► $t = 0$: $i = 0$ (inductor blocks); $t = \infty$: $i = \mathcal{E}/R$
► Find L from decay: $L = Rt/\ln(I_0/I)$
LC Oscillation
$\omega_0 = 1/\sqrt{LC}$; $f = 1/(2\pi\sqrt{LC})$
$q(t) = Q \cos(\omega t)$; $i(t) = -Q\omega \sin(\omega t)$
Energy: $U = Q^2/(2C) = \frac{1}{2}LI^2_{\text{max}}$
★ Damped LRC (series)
★ $\omega' = \sqrt{(1/LC) - R^2/(4L^2)}$
★ Amp decay: $A(t) = A_0 e^{-Rt/(2L)}$; t to fraction f: $t = -(2L/R)\ln(f)$
★ Critical: $R_c = 2\sqrt{L/C}$. $R < R_c$ under; $R > R_c$ over

CH 31 — AC CIRCUITS
$V(t) = V_{\text{max}} \cos \omega t$; $I(t) = I_{\text{max}} \cos(\omega t - \phi)$
$V_{\text{rms}} = V_{\text{max}}/\sqrt{2}$; $I_{\text{rms}} = I_{\text{max}}/\sqrt{2}$
R: in phase, $V_R = IR$
L: V leads 90°; $X_L = \omega L$, $V_L = IX_L$
C: V lags 90°, $X_C = 1/(\omega C)$, $V_C = IX_C$
" <i>ELI the ICE man</i> " — <i>E</i> leads <i>I</i> in <i>L</i> ; <i>I</i> leads <i>E</i> in <i>C</i>
Series LRC
$Z = \sqrt{R^2 + (X_L - X_C)^2}$; $I = V/Z$
$\tan \phi = (X_L - X_C)/R$
Resonance: $X_L = X_C$; $\omega_0 = 1/\sqrt{LC}$; $Z = R$
$P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} \cos \phi = I^2_{\text{rms}} R$

★ Phasor Diagram Tips
★ V_R along I (in phase); V_L straight up (+90°); V_C straight down (−90°). V_{source} = vector sum.
★ $[V_{\text{source}}]^2 = V_R^2 + (V_L - V_C)^2$
★ Transformer (ideal)
★ $V_2/V_1 = N_2/N_1$ (turns ratio)
★ $I_2/I_1 = N_1/N_2$ (inverse, conserves P)
★ $P_1 = P_2 \rightarrow V_1 I_1 = V_2 I_2$ (ideal)
★ Step-up: $N_2 > N_1 \rightarrow V_1, I_1$. Step-down: opposite.

CH 32 — EM WAVES
► $E = E_m \sin(kx - \omega t)$; $B = B_m \sin(kx - \omega t)$
► $E_m/B_m = c$; $E(t)/B(t) = c$
► $c = 1/\sqrt{(\mu_0 \epsilon_0)} = f\lambda = \omega/k$
► $k = 2\pi/\lambda$; $\omega = 2\pi/T = 2\pi f$
► $S = (1/\mu_0)(E \times B)$; $ S = EB/\mu_0 = E^2/(c\mu_0) = cB^2/\mu_0$
► $I = S_{\text{avg}} = E_m B_m/(2\mu_0) = \frac{1}{2}c\epsilon_0 E_m^2 = E_m^2/(2\mu_0 c)$
Rad pressure: Absorb: $P = I/c$. Reflect: $P = 2I/c$
$E \times B \rightarrow$ propagation (RHR). Spectrum (low−high f):
Radio < $\mu W < IR < Vis(400\text{--}700\text{nm}) < UV < X < \gamma$
In medium: $v = c/n$ (n =index of refraction)

★ MAXWELL'S 4 EQUATIONS
1. Gauss (E): $\oint \mathbf{E} \cdot d\mathbf{A} = Q_{\text{encl}}/\epsilon_0$
2. Gauss (B): $\oint \mathbf{B} \cdot d\mathbf{A} = 0$ (no monopoles)
3. Faraday: $\oint \mathcal{E} \cdot d\mathbf{l} = -d\Phi_B/dt$
4. Ampere-Maxwell: $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0(I_C + \epsilon_0 d\Phi_E/dt)$
★ RIGHT-HAND RULE GUIDE
$\mathbf{F} = q\mathbf{v} \times \mathbf{B}$ (chg): fingers <i>v</i> , curl to <i>B</i> , thumb= <i>F</i> (flip if <i>q</i> <0)
B from wire: thumb along <i>I</i> , fingers curl in <i>B</i> dir
Ampere loop: fingers along path, thumb = + <i>I</i> dir
Loop μ: fingers curl with <i>I</i> , thumb = μ dir
EM wave: fingers <i>E</i> , curl to <i>B</i> , thumb = propagation
Lenz: Find <i>d</i> Φ dir, induced <i>B</i> opposes it, then RHR for <i>I</i>
★ COMMON PITFALLS
▲ Φ B in T·m ² (NOT Wb) per prof
▲ RC charging is $1 - e^{-(t/RC)}$; discharging is $e^{-(t/RC)}$. Don't swap.
▲ RL: at $t = 0$ inductor BLOCKS ($i = 0$). Cap at $t = 0$ is wire ($V = 0$).
▲ "Toroid" <i>r</i> is from CENTER (not from wire).
▲ Cap disconnected $\rightarrow$ Q const, <i>E</i> unchanged when <i>d</i> changes. Connected $\rightarrow$ V const.
▲ cos vs sin in Φ: cosθ when θ from normal; sinθ if from plane.
▲ In LRC: $V_C + V_L + V_R \neq V_{\text{source}}$ (alg). Use phasor sum.
▲ In Ampere: <i>r</i> is from current axis, not anywhere else.
▲ Mass spec: <i>V</i> is accelerating voltage, not vel-selector <i>V</i> .
▲ $\tau_{RC} = RC$, $\tau_{RL} = L/R \rightarrow$ DIFFERENT, easy to swap.
▲ "Loop fully inside <i>B</i> " $\rightarrow$ no <i>d</i> Φ $\rightarrow$ <i>I</i> = 0 (only changing flux induces).
▲ Direction of induced <i>I</i> : use Lenz, NOT sign in Faraday's eq.
★ TEST 1 EXTRA HELP (Ch 21-23)
Vector decomposition of <i>E</i> from point <i>q</i> at (<i>x,y</i>):
$E_x = (kq/r^2)\cos\theta = kqx/r^3$; $E_y = (kq/r^2)\sin\theta = kqy/r^3$
$ \mathbf{E} = \sqrt{(E_x^2 + E_y^2 + E_z^2)}$; $\hat{r}$ sign: <i>a</i> (−) $\rightarrow$ use − $\hat{r}$; <i>a</i> (+) $\rightarrow$ use + $\hat{r}$
Solid disk on axis (radius <i>R</i> , σ uniform): $E_x = 2\pi k\sigma[1 - x/\sqrt{x^2 + R^2}]$
Annulus (disk w/ hole, <i>R</i> in & <i>R</i> out): $E_x = 2\pi k\sigma[1/\sqrt{x^2 + R_{\text{in}}^2} - 1/\sqrt{x^2 + R_{\text{out}}^2}]$
Ring V on axis: $V = kQq/\sqrt{z^2 + a^2}$
Disk V on axis: $V = 2\pi k\sigma \int \sqrt{x^2 + a^2} - x]$
Ring max E location: at $z = R/\sqrt{2}$ ($E = 0$ at center, max at $R/\sqrt{2}$)
Charge density for integration:
Linear: $dq = \lambda \, dl$ Surface: $dq = \sigma \, dA$ Volume: $dq = \rho \, dv$
Semicircle: $dq = (Q/\pi a) \cdot a \, d\theta = (Q/\pi) d\theta$
SHM in E-field (charge <i>q</i> oscillating near ring/disk):
$\omega = \sqrt{(kqQ/mR^3)}$; $f = (1/2\pi)\sqrt{(kqQ/mR^3)}$
Conservation of energy (charge moving in E): $K_a + U_a = K_b + U_b$ $\frac{1}{2}mv^2 = q\Delta V$ $v = \sqrt{2(q\Delta V/m)}$
Work btw two points: $W = kq_1 q_2(1/r_1 - 1/r_2)$
ΔV pt charge btw 2 pts: $\Delta V = kQ(1/r_b - 1/r_a)$
E from V (gradient): $E = -dV/dx$; if $V = ar^2 \rightarrow E = -2ar$
Line/cylinder V undefined at ∞ — use ΔV between 2 finite points only

★ TEST 2 EXTRA HELP (Ch 24-27)
★ Force btw cap plates: $F = Q^2/(2\epsilon_0 A) = U/d$ (constant Q)
Cap series/parallel rules:
Series: Q same; V adds ($V_{\text{total}} = V_1 + V_2$); $1/C_{\text{eq}} = \Sigma 1/C_i$
Parallel: V same; Q adds ($Q_{\text{total}} = Q_1 + Q_2$); $C_{\text{eq}} = \Sigma C_i$
Voltage divider (series C): $V_1 = V_{\text{total}}(C_{\text{eq}}/C_1)$
Charge divider (parallel C): $Q_1 = Q_{\text{total}}(C_1/C_{\text{eq}})$
★ Meter rules:
Ammeter: 0 Ω, in SERIES Voltmeter: ∞ Ω, in PARALLEL
★ Kirchhoff loop sign rule:
Walk WITH current thru R: −IR Walk AGAINST current: +IR
Exit + terminal of battery: +ε Exit − terminal: −ε
Battery power breakdown:
Total P = $\mathcal{E}I$ Useful (to R) $P = V_{\text{term}}I = (\mathcal{E} - Ir)I$
Wasted in r: $P_{\text{lost}} = I^2 r$ Energy = Pt
Power absorbed (charging batt): $P = \mathcal{E}I$ (energy goes IN)
Cap work done (Ad): $W = U_{\text{final}} - U_{\text{initial}}$
F = qv×B determinant form:
$F = q \cdot \det[\hat{i} \, \hat{j} \, \hat{k}; v_x \, v_y \, v_z; B_x \, B_y \, B_z]$
Wire perpendicular to B: $F = ILB \sin\theta = 1$
Total charge from N electrons: $q = N \cdot e$ ($e = 1.6 \times 10^{-19}$ C)
Kinematics in B-field (free fall into B): $v^2 = v_0^2 + 2g\Delta y$
V_{xy} potential difference: $V_{xy} = V_x - V_y$

☼ EXAM TRIAGE — KEYWORD → FORMULA

"insulating sphere E in/out" $\rightarrow kQr/R^3$ vs kQ/r^2	"equiv resistance" $\rightarrow$ reduce groups	"period in B-field" $\rightarrow 2\pi m/(qB)$	"LRC critical R" $\rightarrow 2\sqrt{L/C}$
"infinite line/wire E" $\rightarrow \lambda/(2\pi\epsilon_0 r)$	"two R parallel" $\rightarrow R_1 R_2/(R_1 + R_2)$	"cyclotron freq" $\rightarrow qB/(2\pi m)$	"LRC underdamped ω' " $\rightarrow \sqrt{(1/LC - R^2/4L^2)}$
"infinite sheet E" $\rightarrow \sigma/(2\epsilon_0)$	"n equal R parallel" $\rightarrow R/n$	"force wires/length" $\rightarrow \mu_0 I_1 I_2/(2\pi d)$	"amplitude decay LRC" $\rightarrow e^{-Rt/2L}$
"conductor surface E" $\rightarrow \sigma/\epsilon_0$	"power in resistor" $\rightarrow I^2 R = V^2/R$	"toroid B inside" $\rightarrow \mu_0 NI/(2\pi r)$	"X _L " $\rightarrow \omega L$
"point charge E" $\rightarrow kQ/r^2$	"power from battery" $\rightarrow \mathcal{E}I$	"toroid B outside" $\rightarrow 0$	"X _C " $\rightarrow 1/(\omega C)$
"point charge V" $\rightarrow kQ/r$	"battery terminal V" $\rightarrow \mathcal{E} - Ir$	"solenoid B" $\rightarrow \mu_0 nI$	"impedance Z" $\rightarrow \sqrt{R^2 + (X_L - X_C)^2}$
"point charge V" $\rightarrow kQ/r$	"two batteries opposing" $\rightarrow I = \mathcal{E}_1 - \mathcal{E}_2 /\Sigma R$	"moving charge B" $\rightarrow (\mu_0/4\pi)qv \sin\theta/r^2$	"resonance ω' " $\rightarrow 1/\sqrt{LC}$
"ring on axis E" $\rightarrow kqz/(z^2 + a^2)^{3/2}$	"two batteries aiding" $\rightarrow I = (\mathcal{E}_1 + \mathcal{E}_2)/\Sigma R$	"long wire B" $\rightarrow \mu_0 I/(2\pi r)$	"phase angle" $\rightarrow \tan \phi = (X_L - X_C)/R$
"semicircle ±Q" $\rightarrow 4kQ/(\pi a^2)$	"wire R from L, A" $\rightarrow R = \rho L/A$	"inside wire B ($r < R$)" $\rightarrow \mu_0 I r/(2\pi R^2)$	"P avg AC" $\rightarrow I_{\text{rms}}^2 R$
"square alt ±q" $\rightarrow 4\sqrt{2} kq/a^2$	"drift velocity" $\rightarrow I = nqv_d A$	"center loop B" $\rightarrow \mu_0 I/(2R)$	"V _{rms} " $\rightarrow V_{\text{max}}/\sqrt{2}$
"cavity in sphere" $\rightarrow \rho h/(3\epsilon_0)$	"current density J" $\rightarrow I/A$	"on axis loop B" $\rightarrow \mu_0 I R^2/[2(R^2 + z^2)^{3/2}]$	"transformer V" $\rightarrow V_2/V_1 = N_2/N_1$
"flux thru hemisphere" $\rightarrow q/(2\epsilon_0)$	"τ for RC" $\rightarrow RC$	"torque on loop" $\rightarrow \mu \times B = NIAB \sin\theta$	"transformer I" $\rightarrow I_2/I_1 = N_1/N_2$
"Φ pt charge full sphere" $\rightarrow q/\epsilon_0$	"τ for RL" $\rightarrow L/R$	"PE of dipole in B" $\rightarrow -\mu \cdot B$	"EM wave B from E" $\rightarrow B = E/c$
"V from line of charge" $\rightarrow 2k\lambda \ln(r_2/r_1)$	"RC voltage at time t" $\rightarrow V_C = \mathcal{E}(1 - e^{-t/RC})$	"dI/dt in solenoid $\rightarrow \mathcal{E}'" \rightarrow (\mu_0 n r/2)(dI/dt)$	"intensity from E _m " $\rightarrow \frac{1}{2}c\epsilon_0 E_m^2$
"work to move charge" $\rightarrow W = q\Delta V$	"RC current" $\rightarrow (\mathcal{E}/R)e^{-t/RC}$	"charging cap $\rightarrow B'" \rightarrow \mu_0 I_C r/(2\pi R^2)$	"Poynting" $\rightarrow (1/\mu_0)\mathbf{E} \times \mathbf{B}$
"speed gained from V" $\rightarrow v = \sqrt{2qV/m}$	"RC initial (t=0)" $\rightarrow V_C = 0$, $V_R = \mathcal{E}$, $i = \mathcal{E}/R$	"displacement current" $\rightarrow I_D = \epsilon_0 d\Phi_E/dt$	" S " $\rightarrow EB/\mu_0 = E^2/(c\mu_0)$
"E between plates" $\rightarrow V/d$	"RC final (t=∞)" $\rightarrow V_C = \mathcal{E}$, $V_R = 0$, $i = 0$	"rotating coil $\epsilon'" \rightarrow NBA\omega \sin(\omega t)$	"rad pressure absorb" $\rightarrow I/c$
"force on charge in E" $\rightarrow F = qE$	"RC discharge V" $\rightarrow V_0 e^{-t/RC}$	"max EMF rotating" $\rightarrow NBA\omega$	"rad pressure reflect" $\rightarrow 2I/c$
"dipole torque" $\rightarrow -pE \sin\theta$	"time to charge to half" $\rightarrow t = RC \ln 2$	"flux when B(t)" $\rightarrow \mathcal{E} = -A \cos\theta \, dB/dt$	"k from λ" $\rightarrow 2\pi/\lambda$
"dipole U" $\rightarrow -pE \cos\theta$	"time to decay to 1/e" $\rightarrow t = \tau$	"flux when $\theta(t)" \rightarrow \mathcal{E} = BA \sin\theta \, d\theta/dt$	"ω from f" $\rightarrow 2\pi f$
"cap C from area" $\rightarrow \epsilon_0 A/d$	"loop entering B" $\rightarrow I = BLv/R$	"L of solenoid" $\rightarrow \mu_0 N^2 A/\ell$	"c from μ ₀ ε ₀ " $\rightarrow 1/\sqrt{(\mu_0 \epsilon_0)}$
"cap energy" $\rightarrow \frac{1}{2}CV^2 = Q^2/(2C)$	"slide wire EMF" $\rightarrow \mathcal{E} = BLv$	"torque on loop" $\rightarrow \mu \times B = NIAB \sin\theta$	"E and B in phase" $\rightarrow$ both $\sin(kx - \omega t)$
"cap energy density" $\rightarrow \frac{1}{2}\epsilon_0 E^2$	"force to push rod" $\rightarrow B^2 L^2 v/R$	"inductor energy" $\rightarrow \frac{1}{2}LI^2$	"Maxwell $\oint \mathcal{E} \cdot d\mathbf{A}" \rightarrow 0$
"cap with dielectric" $\rightarrow C = \kappa\epsilon_0 A/d$	"power dissipated rod" $\rightarrow (B^2 L^2)/2R$	"u_B (B-field density)" $\rightarrow B^2/(2\mu_0)$	"Maxwell $\oint B \cdot d\mathbf{A}" \rightarrow 0$
"cap disconn., move plates" $\rightarrow Q$ const, U ad	"undeflected ion" $\rightarrow v = E/B$	"RL energizing" $\rightarrow (\mathcal{E}/R)[1 - e^{-Rt/L}]$	"Maxwell $\oint \mathcal{E} \cdot d\mathbf{l}" \rightarrow -d\Phi_B/dt$
"cap stays connected" $\rightarrow V$ const, U ±1/d	"mass spec m/q" $\rightarrow B^2 R^2/(2V)$	"RL decay" $\rightarrow I_0 e^{-Rt/L}$	"Maxwell $\oint B \cdot d\mathbf{l}" \rightarrow \mu_0(I_C + \epsilon_0 d\Phi_E/dt)$
"caps in series" $\rightarrow 1/C_{\text{eq}} = \Sigma 1/C_i$	"speed after accel V" $\rightarrow \sqrt{2qV/m}$	"RL find L" $\rightarrow L = Rt/\ln(I_0/I)$	"speed in medium" $\rightarrow c/n$
"caps in parallel" $\rightarrow C_{\text{eq}} = \Sigma C_i$	"radius circ motion B" $\rightarrow r = mv/(qB)$	"LC freq" $\rightarrow 1/(2\pi\sqrt{LC})$	" $\epsilon \times \lambda" \rightarrow c$ (or <i>v</i> in medium)
"two C in series" $\rightarrow C_1 C_2/(C_1 + C_2)$		"LC $\omega"$ $\rightarrow 1/\sqrt{LC}$	"E density u _E " $\rightarrow \frac{1}{2}\epsilon_0 E^2$

★★★ FINAL EXAM — STARRED PROBLEM TYPES ★★★
△ "NEVER USE WEBERS" — keep magnetic flux as T·m², NOT Wb (1 Wb = 1 T·m²)

① Insulating Sphere — E inside/outside

► Uniform Q in sphere radius R. **Inside (r<R):** E = kQr/R³
Outside (r≥R): E = kQ/r²

Worked: E=760 N/C at r=R/4 (inside): kQ(R/4)/R³ = kQ/(4R²) = 760 → kQ/R² = 3040.
At r=4R (outside): E = kQ/(4R)² = kQ/(16R²) = 3040/16 = **190 N/C**
If asked at r=R: both formulas give E=kQ/R²

② Equivalent Resistance

► **Series:** R_{eq} = R₁+R₂+R₃...
► **Parallel:** 1/R_{eq} = 1/R₁+1/R₂+...
► **Two parallel:** R_{eq} = R₁R₂/(R₁+R₂)

► **Method:** Reduce innermost groups first → simplify outward → I_{total} = ε/R_{eq} → back-substitute V=IR for each branch.

► **Same R in parallel** (n of them): R_{eq} = R/n

③ RC Circuit (timing)

► **Charging:** q(t) = Q_{max}[1 - e^{-t/RC}], Q_{max}=Cε
i(t) = (ε/R)e^{-t/RC}
V_C(t) = ε(1 - e^{-t/RC}); V_R = εe^{-t/RC}

► **At t=0:** V_C=0, V_R=ε(MAX), i=ε/R(MAX), q=0
At t=∞: V_C=ε(MAX), V_R=0, i=0, q=Q_{max}
τ = RC (63% of final)

► **Discharging:** q=Q₀e^{-t/RC}; i=(Q₀/RC)e^{-t/RC}; V=(Q₀/C)e^{-t/RC}
Correction: exponent is -t/RC (not -ε/RC)

④ B from Moving Charge

► **B = (μ₀/4n)(qv×r̂)/r²**
|B| = (μ₀/4n)(qv sinθ)/r²
► θ = angle between v and r. RHR for direction.

⑤ Long Straight Wire

► **B = μ₀I/(2πr)** (outside, dist r from axis)
► **Direction:** RHR — thumb along I, fingers curl in B direction
► Inside (r<R): B = μ₀Ir/(2πR²)

⑥ Ampere's Law (Multiple Currents)

► **∮B·dl = μ₀I_{encl}**
► **Sign rule:** Curl right hand along path direction. Thumb = + current direction.
★ **CCW path** → + out of page
★ **CW path** → + into page

► **I_{encl}** = sum with signs: I₁(±) + I₂(±) + I₃(±)
► **Then:** B(2πr) = μ₀I_{encl} → B = μ₀I_{encl}/(2πr)

⑦ Toroid B-field

► **B = μ₀NI/(2πr)** (inside, r = dist from CENTER axis)
► N = total turns. Outside: B = 0

△ Flux Φ_B = BA in T·m², not Wb

⑧ Faraday's Law of Induction

► **ε = -dΦ_B/dt** Φ_B = BA cosθ
► **If B(t) varies, θ const:** ε = -A cosθ·(dB/dt)
► **If θ(t) varies, B const:** ε = BA sinθ·(dθ/dt)
► **Both vary:** ε = -A[cosθ·dB/dt - B sinθ·dθ/dt]

► **Rotating coil:** θ=ωt → ε = NBAω sin(ωt); ε_{max} = NBAω
► **N loops:** multiply ε by N

⑨ Slide-Wire Generator

► **ε = vBL** (motional EMF)
► v = rod speed, B = field, L = rod length perp to v & B
► **Induced current:** I = ε/R = vBL/R
Force to push: F = BIL = B²L²v/R
Power: P = I²R = (vBL)²/R

⑩ RL Circuit

► **Energizing:** i(t) = I_{max}[1 - e^{-Rt/L}]; I_{max}=ε/R
Decay: i(t) = I₀e^{-Rt/L}

► **τ = L/R** | V_L(t) = ε·e^{-Rt/L} (energizing)
► **At t=0:** i=0 (inductor blocks). t=∞: i=ε/R

⑪ EM Waves — Full Set

► **E(x,t) = E_m sin(kx - ωt)**
B(x,t) = B_m sin(kx - ωt)

► **Amplitude ratio:** E_m/B_m = c
Instant ratio: E(t)/B(t) = c

► **Wave speed:** c = 1/√(μ₀ε₀) = 3×10⁸ m/s
Wave number: k = 2π/λ [rad/m]
Angular freq: ω = 2π/T = 2πf [rad/s]
c = fλ = ω/k

► **Poynting:** S = (1/μ₀)(E × B) [W/m²]
|S| = EB/μ₀ = E²/(cμ₀) = cB²/μ₀
Avg I: I = E_mB_m/(2μ₀) = ½cε₀E_m²
► **Direction:** E × B = propagation (RHR)

CH 21-24 CONDENSED (1 Q)

Coulomb & E-fields

F = kq₁q₂/r²; **E = kQ/r²** (point)

Inf line: E=λ/(2πε₀r)=2kλ/r

Inf sheet: E=σ/(2ε₀); **Conductor:** E=σ/ε₀

Ring axis: E=kqz/(z²+a²)^{3/2}

★ Square ctr alt±q: E=4v/2.kq/a²

★ Semicircle ½±q: E=4kQ/(πa²)

★ Zero-E pt (2 like q): x=s/(1+√(q₂/q₁))

★ Two ∥ sheets same sign: between=|σ_A-σ_B|(2ε₀); outside=(σ_A+σ_B)/(2ε₀). Opp sign: swap.

Dipole

τ=p×E; |τ|=pE sinθ

U=-p·E=-pE cosθ; F_{net}=0 unif

Gauss

Φ=∫E·dA=Q_{encl}/ε₀

Inside unif sphere: E=kQr/R³=pr/(3ε₀)

Inside unif cyl: E=pr/(2ε₀)

★ Cavity in sphere: E=ρh/(3ε₀) UNIFORM

Potential

V=kQ/r; U=qV; ΔV=-∫E·dl; E=-dV/dx

W_{field}=qΔV=-ΔU; ½mv²=qΔV (rest)

Plates: E=V/d. Line: ΔV=2kλ ln(r₀/r)

Capacitors

Q=CV; C=ε₀A/d (∥-plate); κε₀A/d w/diel

Series: 1/C_{eq}=Σ1/C_i; Par: C_{eq}=ΣC_i

U=½CV²=Q²/(2C)=½QV; u=½ε₀E²

★ Disconn (Q const): U_{new}/U_{old}=d_{new}/d_{old}

Connected (V const): U_{new}/U_{old}=d_{old}/d_{new}

CH 25 — EMF & CIRCUITS

I=ε/(R+r); V_{term}=ε-Ir (discharge)

V=IR; **P=IV=I²R=V²/R**; W=εIt

★ Aiding batt: I=(ε₁+ε₂)/ΣR; Opp: I=|ε₁-ε₂|/ΣR (dir = bigger ε wins)

Current Density & Resistivity

I = nqv_dA (n=charge/vol, v_d=drift)

J = I/A = nqv_d [A/m²]

R = ρL/A; ρ=resistivity [Ω·m]; σ=1/ρ

E = ρJ (microscopic Ohm's)

Temp: ρ(T)=ρ₀[1+α(T-T₀)]

CH 26 — KIRCHHOFF & RC

Loop: ΣV=0; Junction: ΣI_{in}=ΣI_{out}

R-series: R_{eq}=ΣR_i; R-par: 1/R_{eq}=Σ1/R_i

P_{dissipated} by R = I²R = V²/R

★ Find Time Algebra (RC/RL)

★ **Charging V_C=αε:** t = -RC·ln(1-α)
α=½ → t=RC·ln(2)≈0.693RC

★ **Decay to fraction f of I₀:** t = -τ·ln(f) (works RC and RL with τ=RC or L/R)

★ **RL energize to fraction f of I_{max}:** t = -(L/R)·ln(1-f)

★ **τ table:** 1τ→63%, 2τ→86%, 3τ→95%, 5τ→99.3%

★ Energy-Storage Cheat

Element	Energy	Density
Capacitor	½CV ² =Q ² /2C	u=½ε ₀ E ²
Inductor	½LI ²	u=B ² /(2μ ₀)
EM Wave	—	u=ε ₀ E ² =B ² /μ ₀

In EM wave: u_E = u_B (equal split)

★ Math Quick-Ref

Derivatives: d/dt(e^{at})=ae^{at}; d/dt(sin ωt)=ω cos ωt; d/dt(cos ωt)=-ω sin ωt

Vectors: A·B=AB cosθ (scalar); A×B=AB sinθ (vector ⊥ both, RHR)

Geometry: Circle area=πr²; Sphere area=4πr², vol=(4/3)πr³; Cylinder side=2πrL; Annulus=π(R₂²-R₁²)

Quadratic: x = (-b±√(b²-4ac))/(2a)

Logs: ln(e^x)=x; ln(ab)=ln a+ln b; ln(a/b)=ln a-ln b; ln(1)=0; ln(2)≈0.693

Trig: sin²+cos²=1; sin(2θ)=2sinθcosθ; cos(2θ)=cos²θ-sin²θ

Common angles: sin/cos at 0°=0/1; 30°=½/(√3/2); 45°=√2/2; 60°=(√3/2)/½; 90°=1/0

CONSTANTS & CONVERSIONS

k = 1/(4πε₀) = **8.99×10⁹ N·m²/C²**
ε₀ = **8.854×10⁻¹² C²/N·m²**
μ₀ = **4π×10⁻⁷ = 1.257×10⁻⁶ T·m/A**
μ₀/(4π) = **10⁻⁷ T·m/A**
c = 1/√(μ₀ε₀) = **2.998×10⁸ m/s**
e = **1.602×10⁻¹⁹ C**
m_p = **9.109×10⁻³¹ kg** = 0.511 MeV/c²
m_p = **1.673×10⁻²⁷ kg** = 938 MeV/c²
g = **9.81 m/s²**
1 eV = **1.602×10⁻¹⁹ J**
1 N/C = **1 V/m**
1 Wb = **1 T·m²** △ DON'T USE Wb
1 H = **1 V·s/A = 1 T·m²/A**
1 Hz = **1/s**; 1 rad ≈ 57.3°
1 G = **10⁻⁴ T**; 1 mT = **10⁻³ T**

△ UNITS REFERENCE						
Quantity	Sym	Unit				
Charge	q, Q	C = A·s				
E-field	E	N/C = V/m				
Potential	V	V = J/C				
Energy	U	J = C·V				
Capacitance	C	F = C/V				
Current	I	A = C/s				
Curr density	J	A/m ²				
Resistance	R	Ω = V/A				
Resistivity	ρ	Ω·m				
Power	P	W = J/s				
EMF	ε	V				
B-field	B	T = N/(A·m)				
Mag flux Δ	Φ, B	T·m ² (NOT Wb)				
Inductance	L	H = V·s/A				
Reactance	X _L , X _C	Ω				
Impedance	Z	Ω				
Mag moment	μ	A·m ² = J/T				
Frequency	f	Hz = 1/s				
Angular freq	ω	rad/s				
Wavelength	λ	m				
Wave number	k	rad/m				
Period	T	s				
Energy dens	u	J/m ³				
Intensity	I, S	W/m ²				
Rad pressure	P _{rad}	Pa = N/m ²				
Linear chg	λ	C/m				
Surface chg	σ	C/m ²				
Volume chg	ρ	C/m ³				
Permittivity	ε ₀	F/m				
Permeability	μ ₀	H/m				
Dipole moment	p	C·m				
Drift vel	v _d	m/s				
Conductivity	σ	S/m				
SI	p	n	μ	m	c	k M G
10 [^]	-12	-9	-6	-3	-2	3 6 9

★ WORKED PRACTICE EXAMPLES & DECISION TREE

WE-1 ► RC Charging

R=10 kΩ, C=100 μF, ε=12 V. Find t for V_C=8 V.

τ = RC = **1.0 s**; **1.12 s** (1 - e^{-t/τ}) → e^{-t/τ} = 1/3
t = -τ ln(1/3) = **8.10 s**

WE-2 ► Long Wire B

I=5 A, r=2 cm.

B = μ₀I/(2πr) = (10⁻⁷·5)/(0.02) = **5×10⁻⁵ T**
Sanity: ≈ Earth's B (50 μT) ✓

WE-3 ► Mass Spec

V=1000 V, B=0.5 T, R=0.05 m.

m/q = B²R²/(2V) = (0.25)/(0.0025)/2000 = **3.13×10⁻⁷ kg/C**
Singly-ionized: m ≈ **5×10⁻²⁶ kg** (~30 amu)

WE-4 ► Slide-Wire

B=0.4 T, L=0.3 m, v=2 m/s, R=5 Ω.

ε = BLv = **0.24 V**; I = **48 mA**
F_{push} = B²L²v/R = **5.76 mW**; P = εI = **11.5 mW**

WE-5 ► Rotating Coil

N=50, A=0.01 m², B=0.2 T, ω=377 rad/s.

ε_{max} = NBAω = **37.7 V (peak)**
V_{rms} = 37.7/√2 = **26.7 V**

WE-6 ► Inductor Energize

ε=24 V, R=12 Ω, L=0.6 H. Find i at t=0.05 s.

τ = L/R = 0.05 s (one tau!); I_{max} = 2 A
i(t) = I_{max}(1 - e^{-t/τ}) = **1.26 A**

WE-7 ► LRC Resonance + EM Wave

L=0.1 H, C=10 μF; ω₀ = 1/√(LC) = **1000 rad/s**;
f₀ = **159 Hz**
EM wave E_m=100 V/m; B_m = E/c = **3.33×10⁻⁷ T**; I = ½cε₀E_m² = **13.3 W/m²**

★ COMMON PRODUCTS

μ₀/(4π) = 10⁻⁷ T·m/A | Z₀ = 1/(cε₀) ≈ **377 Ω**
1/√2 ≈ 0.707; e⁻¹ ≈ 0.368; π² ≈ 9.87
ln(2) ≈ 0.693; ln(10) ≈ 2.303; ln(3) ≈ 1.099

★ DECISION TREE — Pick Formula Fast

Charge in B-field, find r → r = mv/(qB)

Two parallel wires → F/L = μ₀I₁I₂/(2πd)

Loop moves through B → ε = BLv

B(t) changing, loop fixed → ε = -A(dB/dt)

Loop rotating in const B → ε = NBAω sin(ωt)

Inductor at t=0: i=0 (open) | at t=∞: i=ε/R

Capacitor at t=0: V=0 (wire) | at t=∞: i=0

AC find Z → √(R²+(X_L-X_C)²)

AC resonance ω → 1/√(LC)

EM wave: E given, find B → B = E/c

EM wave intensity → ½cε₀E_m²

Cap connected, d↑ → V const, U↓

Cap disconnected, d↑ → Q const, U↑

Wire R from material → R = ρL/A

Power dissipated → I²R = V²/R